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Media coverage of lobbies in Spanish digital generalist press

Abstract: Media coverage is one of the most conditioning dimensions affecting the influence capacity of lobbies, as it enables them to bolster their stance, keep their supporters informed, and communicate their viewpoints to decision-makers. The present study focuses on examining the media coverage of lobbies on the websites of the leading generalist newspapers in Spain from January 2013 to December 2023. A total of 13,431 news articles related to the topic were collected, published online on the seven most-read generalist newspapers in the country. The analysis of the results reveals that most news articles address political-economic interests. It is also found that the sentiment expressed in their content is predominantly negative. Interestingly, more than a third of the news articles are published as a result of an indirect lobbying strategy, and these publications express a significantly more positive sentiment compared to the rest. This suggests that while the general coverage tends to reflect skepticism towards lobbies, news articles published as a consequence of indirect lobbying strategies can present a more favorable perspective about these organizations.

Keywords: lobbying, media, generalist press, media coverage, influence.

Word count: 7895 (excluding the parts indicated in the journal's instructions)

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1. Introduction

Currently, the structure of modern societies is more complex than ever (Carpani, 2024; Petrov, 2023). This reality has led to the emergence of numerous associations and groups representing specific social sectors with shared interests and goals (Castillo, 2011; Junk, 2020). For decades (Basso, 1983) up to the present (Colli and Adriaensen, 2020; Dür and Mateo, 2024), many scholars have emphasized the importance of these entities in shaping the civil sphere, as they facilitate access and contact with political institutions (Judd, 2023). When these organizations have a stable structure, endure over time, and aim to influence political decision-making, they are considered pressure groups or lobbies (Castillo, 2011).

The fact that lobbies seek to influence the policy-making process is one of the most crucial aspects of their operational functioning, generating interest in the scientific literature. To assess the capacity of these organizations to influence, three interdependent groups of conditioning dimensions must be considered (Serna-Ortega et al., 2025).

The first group includes five dimensions that are independent and unchanging regarding the lobby's context and objectives. The most apparent of these is financial strength, as economic resources determine the actions that can be undertaken (Almansa-Martínez et al., 2022; Stevens and De Bruycker, 2020). Another dimension is the ability to mobilize supporters, as demonstrations of support, both physical and digital, are highly valued in democratic systems (Turnbull-Dugarte et al., 2022; Vathi and Trandafoiu, 2023; Walker, 2012). Moreover, coalitions often form between groups with shared interests, enhancing collective influence and maximizing the impact of their efforts (Grose et al., 2022; Heermann, 2024; Junk, 2020). Aligning with social norms also increases the likelihood of gaining support from both authorities and the public (Ihlen and Raknes, 2020; Raknes and Ihlen, 2020; Scott, 2014), thereby improving the organization's long-term reputation and credibility (Barron and Skountridaki, 2022; Bernhagen et al., 2022). Finally, access to public authorities is both a goal and a key factor of influence, necessitating the identification of suitable intermediaries and the adjustment of strategies according to power dynamics (Junk et al., 2022; Hirsch et al., 2023).

The second group consists of two dimensions that depend on and vary with the context and objectives. One of them is the stage of the policy-making process where influence is sought (Judd, 2023; Murphy, 2023; Serna-Ortega et al., 2025). According to Jordan and Adelle (2012), these stages include problem emergence,

agenda-setting, consideration of policy options, decision-making, implementation, and evaluation of effects. Kraft and Furlong (2015) propose a simplified version that merges the first two stages. Communication strategies also play a significant role, as their effectiveness directly impacts influence capacity. These strategies can be direct, targeting policymakers (Awad, 2024; Lock and Davidson, 2024), or indirect, mobilizing public opinion to exert pressure (Moreno-Cabanillas et al., 2024; Olejnik, 2023).

The third and final group comprises two conditioning dimensions that are ultimately beyond the lobby's direct control. Consequently, their determinism on the capacity to influence is tangential. The first one is public perception, which, although not extensively studied, allows for certain conclusions to be drawn based on the research that has addressed the topic in various ways (Dür and Mateo, 2014; Junk and Rasmussen, 2024; Rasmussen and Willems, 2022). A favorable perception enhances the legitimacy of an organization's demands and facilitates support from both the public and the media (Rasmussen et al., 2018). Conversely, a negative perception can weaken the persuasive capacity with policymakers, who tend to avoid controversy and prefer to align with majority opinion (Lax and Phillips, 2012; Shapiro, 2011). Media coverage is the other dimension within this group, as it amplifies or limits the reach of the group's initiatives (Moreno-Cabanillas et al., 2024) and, to some extent, determines their influence capacity and the effectiveness of their strategic proposals, potentially being decisive in the interactions between citizens, lobbies, and decision-makers (De Bruycker and Beyers, 2015; Sobbrío, 2011). Its significance lies in the fact that it confers political dimension to claims (Neal, 2008). Overall, lobbies strive to gain visibility in the media to strengthen their position, keep their supporters informed, and convey their perspectives to policymakers (De Bruycker and Beyers, 2015; Luo et al., 2021; Stevens and De Bruycker, 2020). Thus, media presence serves a threefold function for these organizations, with effects that can manifest both in the short and long term.

This study focuses on exploring this final conditioning dimension: media coverage. The general objective (**GO**) is to examine the media coverage of lobbies on the websites of the leading generalist newspapers in Spain from January 2013 to December 2023. Based on this GO, five specific objectives (**SOs**) are defined:

- **SO1.** Analyze the temporal and website distribution of the news articles.
- **SO2.** Investigate the most frequent terms used in the news articles.
- **SO3.** Identify and classify the categories of interests addressed in the news articles.
- **SO4.** Study the sentiment expressed in the content of the news articles.
- **SO5.** Explore whether the publication of the news articles responds to indirect lobbying strategies or not, as well as the explanatory variables.

The structure of the study follows a conventional framework. First, it reviews the existing scientific literature on the topic, followed by a description of the methodological approach designed to achieve the GO and the SOs. Next, the results are presented in five sections, each corresponding to a SO. Finally, the findings are discussed in relation to previous research and summarized in the conclusions, where the implications, limitations, and potential future research directions are also outlined.

2. Theoretical background

The review of the scientific literature indicates that the level of media coverage of lobbies is determined by the actors involved and the institutional contexts in which they operate (De Bruycker and Beyers, 2015; Moreno-Cabanillas et al., 2024; Stevens and De Bruycker, 2020; Tresch and Fischer, 2015). The overall trend suggests that the media often tend to favor certain interests, especially those related to the economic and business sectors, which are often associated with groups that have organizational characteristics linked to private interests (Baumgartner and Leech, 2001; Castillo-Esparcia et al., 2017; Lowery and Gray, 2004).

Generally, most research focuses on determining which organizations gain the most visibility in the media (Baumgartner and Leech, 2001; De Bruycker and Beyers, 2015; Lowery and Gray, 2004; Moreno-Cabanillas et al., 2024). Some studies have also attempted to correlate media coverage of specific interests with the number of groups representing them (Bernhagen and Trani, 2012; Binderkrantz, 2005), or with the number of people who share those concerns (Schlozman and Tierney, 1986). A novel aspect of the present study lies in the intention to detect discrepancies in media coverage based on interests addressed, analyzed in an integral and joint manner. Rather than focusing on measuring coverage on a particular topic, the goal is to conduct a comparative assessment, identifying trends and variations. This way of proceeding can be interpreted through the lens of agenda-setting theory. According to McCombs and Shaw (1972), media outlets play a crucial role in determining which issues gain public attention and how they are framed.

Going a step further, other authors have explored how the media can function as lobbies themselves, selecting content that favors their interests and excluding those that do not. In this way, they become intermediaries of their own agenda, which forces organizations to identify media outlets with editorial lines compatible with their objectives. Consequently, from a theoretical perspective, the assumption that the media act independently creates a dual scenario regarding the impetus to represent an organization or its interests. This representation can result from an indirect lobbying strategy or from media coverage that arises from an autonomous decision. It is crucial to make this distinction, as the aim of lobbies is not merely to gain visibility but to achieve specific objectives through it (Mykkänen and Ikonen, 2019). In this regard, Castillo (2011) argues that for media presence to be effective, it must be accompanied by positive connotations and occur regularly. This duality can be better understood within the framework of the mediated public sphere (Habermas, 2006) and agenda-building theory (Cobb and Elder, 1971), both of which emphasize the interdependence between media, political actors, and the public in shaping issue salience. In this context, lobbying efforts often aim to influence the media agenda, using visibility as an indirect mechanism to steer public debate. At the same time, media organizations exercise their own power to construct reality, prioritizing issues according to professional routines, economic pressures, or ideological orientations (Cook, 2005; Shoemaker and Reese, 2013).

From this duality in the causes of publication derives another novel aspect of this study, namely the exploration of whether a publication stems from an indirect lobbying strategy or not, as well as the explanatory variables. This exploration is novel insofar as it has not been previously conducted and represents a limitation identified in studies such as Moreno-Cabanillas et al. (2024). The variables selected for analyzing their potential explanatory power are justified based on the literature review presented and the formulation of the SOs. In this regard, the first of these is website of publication, as media outlets act as actors with their own agendas (McCombs and Valenzuela, 2021), selecting content that favors certain interests and, at times, excluding others (Castillo-Esparcia et al., 2017; Hamborg et al., 2019). The ideology should also be considered in this context (Nawaz et al., 2013). Closely related, variables concerning the interests addressed are also included, to overcome the methodological issues of previous studies (Baumgartner and Leech, 2001; Moreno-Cabanillas et al., 2024) that emphasize differences based on varying interests but do not explore potential causes in depth. Finally, considering their connection with the emotional framing of the content (Gorostiza-Cerviño et al., 2024), variables related to sentiment are considered. A highly relevant and insightful reference on this topic is the study by Puschmann and Powell (2018).

Research in the Spanish context, which is the geographic focus of the present study, is relatively scarce. An analysis conducted by Castillo-Esparcia et al. (2017) of over a thousand news articles published between 2014 and 2016 revealed that economic lobbies are the most frequently mentioned. Moreover, the authors highlighted that economic crises tend to increase the activity of these lobbies in regulatory processes. Similarly, Moreno-Cabanillas et al. (2024) examined 2,170 news articles in digital media between 2017 and 2022, finding that generalist newspapers pay greater attention to interest groups, with a particular emphasis on employer organizations and public institutions.

3. Materials and methods

The proposed methodology adopts a quantitative approach, as it is considered the most suitable for the data collection, systematization, and analysis procedures. The scope is descriptive-inferential.

3.1. Data collection

To collect the news articles, consultations of secondary databases were conducted. They were collected from the digital repositories of newspapers and the MyNews platform. The effectiveness of these tools has been confirmed in previous studies (Arumi and Sukmasetya, 2020; Shreesha et al., 2018). Relevant news articles were identified using the following keywords: "lobby", "lobbies", "lobbying", "pressure groups", and "interest groups". This approach is considered the most representative and is supported by previous studies on media coverage of lobbies in the Spanish context (Castillo-Esparcia et al., 2017; Moreno-Cabanillas et al., 2024). It is true that this method inevitably excludes certain publications; however, it is regarded as the most suitable option, since any alternative such as conducting individualized searches would be practically unfeasible and would still rely on keywords and manual supervision for proper filtering. Data collection occurred between July 2023 and March 2024. Since this procedure was carried out at a single point in time, the research is classified as non-experimental and cross-sectional.

This approach to data collection initially involves determining the websites included in the study. The websites of the seven Spanish generalist newspapers with the highest readership, according to data from the General Media Framework in Spain (AIMC, 2024), were selected: elpais.com, elmundo.es, lavanguardia.com, abc.es, 20minutos.es, elcorreo.com, and elperiodico.com.

Once the process was completed, 13,431 news articles about lobbies published between January 1, 2013, and December 31, 2023, were identified across the seven mentioned websites.

3.2. Data systematization

After collecting the news articles, it is necessary to define the research variables, determine their operationalization, and code them. For this purpose, content analysis is employed, a technique particularly suitable for studying journalistic material. **Table 1** is provided to define the variables and how they relate to each SO, including their operationalization:

[Table 1]

Table 1: Identification and operationalization of variables.

SO	Variable	Typology	Operationalization
SO1	Date	Temporal	DD/MM/YYYY
SO1	Website of publication	Categorical	7 websites
SO1	Newspaper's ideology	Categorical	Left/Center/Right
SO2	Length	Numerical	Number of characters
SO2	Terms used	Textual	Frequently used terms
SO2	Images and/or videos	Binary (Categorical)	Yes/No
SO3	Macro-category of interests	Categorical	Political-Economic/Mixed/Social
SO3	Category of interests	Categorical	10 categories of interests
SO4	Sentiment score	Numerical	Value on the scale -100 to 100
SO4	Sentiment	Categorical	Ranges on the scale -100 to 100
SO4	Subjectivity	Binary (Categorical)	Objective/Subjective

SO4	Irony	Binary (Categorical)	Non-ironic/Ironic
SO5	Cause of publication	Binary (Categorical)	Indirect lobbying strategy/Autonomous decision

In terms of justification, the selection of variables is grounded in previous research and aims to address existing gaps in the scientific literature:

- SO1 and SO2 variables: their inclusion is justified insofar as they serve to provide a general contextualization of the coverage. These variables are commonly used in studies on media coverage of lobbying activities (Castillo-Esparcia et al., 2017; Moreno-Cabanillas et al., 2024) and in the vast majority of studies on media coverage in general.
- SO3 variables: their inclusion is justified by the need to incorporate a contrastive element that makes it possible to link the coverage to the lobbies. According to Serna-Ortega et al. (2025), in the case of lobbying activities, such differentiation usually depends on either the organizational nature or the interests. Given that most prior research focuses on organizational nature, the present study is based on categories of interests.
- SO4 variables: their inclusion is justified by their potential to provide objective analytical insights from the content. Although sentiment does not measure stance in a strict sense, it offers a reliable proxy for understanding the emotional framing of a text, as applied by Gorostiza-Cerviño et al. (2024). This approach is further supported by Castillo (2011) and Mykkänen and Ikonen (2019), who argue that it is not only crucial to ensure media presence, but that such coverage should be, to some extent, favorable in order to contribute to achieving the organization's strategic objectives, making sentiment a pertinent indicator in this regard.
- SO5 variable: its inclusion is justified by its novelty in the field of research. The duality established based on the cause of publication is an innovative idea grounded, as noted above, in different communication theories (Cobb and Elder, 1971; Habermas, 2006; McCombs and Shaw, 1972).

In addition to presenting the variables and their justification, it is necessary to explain how their practical coding is carried out. The variables corresponding to the date, the website of publication, the length, the terms used, and the presence of images and/or videos are self-evident, as they are extracted directly from the news article. In contrast, the other variables require a detailed explanation of the methodological tools used for their coding.

The coding of the newspaper's ideology variable is based on the Ochiai coefficient from Guerrero-Solé's research (2021) and is enriched with additional information from other authors (López-Rabadán and Casero-Ripollés, 2012; Sádaba-García and Rodríguez-Virgili, 2007). According to this data, elmundo.es, abc.es, and elcorreo.com have a conservative inclination (right); elpais.com, lavanguardia.com, and elperiodico.com are considered progressive (left); while 20minutos.es is located in the center of the ideological spectrum (center). Nevertheless, this is open to debate and should be understood merely as an operational approximation for the purposes of the study.

The systematization of the variable related to category of interests requires two clarifications. The first concerns the operationalization of the categories: they are based on the EU Transparency Register, which details 40 types of possible interests associated with the activities of lobbies. To facilitate analysis and improve clarity, these 40 types of interests have been grouped into ten main categories.

- **ID 01. Politics and foreign affairs:** institutional affairs, enlargement, foreign and security policy, external relations, European Neighborhood Policy.
- **ID 02. Economy, business, and trade:** business and industry, economy, finance and euro, banking and finance, trade, competition, single market.

- **ID 03. Society and social affairs:** employment and social affairs, education and training, youth, culture, culture and media, sports.
- **ID 04. Research and innovation:** research and innovation, digital economy and society, trans-European networks.
- **ID 05. Environment and energy:** environment, energy, climate action, maritime affairs and fisheries.
- **ID 06. Justice and rights:** justice and fundamental rights, borders and security, fraud prevention, migration and asylum.
- **ID 07. Health and food safety:** public health, consumers, food safety.
- **ID 08. Development and cooperation:** international cooperation and development, humanitarian aid and civil protection, regional policy.
- **ID 09. Primary sector:** agriculture and rural development, fisheries.
- **ID 10. Governance and administration:** customs, taxation, budget, communication.

The second concerns the coding process itself. Each category of interests is associated with a set of between 100 and 120 keywords. To determine the most suitable category for each news article, a program developed in R is used. The program operates simply: it counts the number of words in each news article that match the keyword lists of the categories and assigns the article to the category with the highest number of matches. The program is designed to identify matches even if the keywords vary by up to two letters. Additionally, for keywords composed of multiple words, the software considers a match valid if at least 51% of the words match. The validity of the coding for this variable is ensured through manual supervision by the researchers, which allows for the correction of any misclassifications by the automated procedure and guarantees the reliability of the data used in the analyses. News articles that the program is unable to code due to a lack of keyword matches are also manually coded.

In some sections of the results, a broader classification is used for the categories of interests. This classification groups them into three macro-categories: political-economic interests (ID 01, 02, 10), mixed interests (ID 05, 09), and social interests (ID 03, 04, 06, 07, 08).

The content analysis also includes variables related to sentiment, subjectivity, and irony. The coding of these variables is conducted using the MeaningCloud software. Sentiment is coded in two ways: first, using its original value on a scale ranging from -100 to 100, and second, by establishing five categories: very negative (-100 to -80), negative (-80 to -20), neutral (-20 to 20), positive (20 to 80), and very positive (80 to 100).

The last variable is the cause of publication. Two main scenarios are distinguished: in the first, the publication of the news article is the result of an indirect lobbying strategy; in the second, the media publishes it autonomously. The coding of this variable is done manually, evaluating five aspects: the sources of information, the variety of perspectives, the supplementary material to the news, the coverage patterns, and the media's self-declaration. They are based on previous studies (Bernhagen and Trani, 2012; Binderkrantz, 2005; Castillo-Esparcia et al., 2017; De Bruycker and Beyers, 2015; Moreno-Cabanillas et al., 2024; Mykkänen and Ikonen, 2019) and are enriched by the researchers' judgment.

The variables corresponding to SO1 and SO2 are coded from the entirety of the news articles. However, for the variables corresponding to SO3, SO4, and SO5, only a sample of news articles is coded. With a confidence level of 95% and a margin of error of 5%, it is determined that a sample of 380 news articles is required. The selection of these 380 news articles was made using random stratified sampling by website of publication, employing a program developed in R. The representativeness of this procedure is based on the advantages and limitations discussed by Otzen and Manterola (2017). According to these authors, this approach ensures that the sample adequately represents the population, provides more precise estimates,

and maintains a close resemblance to the characteristics of the population. A detailed justification for why the variables from these SOs are coded in a sample of news articles is provided below:

- SO3 variables: as mentioned, these variables are initially coded using a custom-developed program. However, the program does not allow for full automation, which would make it impossible to supervise the process. Considering both the proportion of articles that the program fails to code and those requiring manual intervention to correct the coding, the program achieves an accuracy rate of approximately 78%. Using a sample, however, is much more representative, as the coding can be supervised, and the representativeness reaches 95%. This representativeness derives from the manual supervision of the coding process and should be understood in light of its statistical validity, as previously justified based on Otzen and Manterola (2017). It is also important to understand that, using the sample, representativeness is calculated based on the population, not on the coding process.
- SO4 variables: in this case, the variables are coded only in the news articles from the sample because the subsequent analyses involve exploring dependencies between these variables and those from SO3 and SO5, which require a common analytical sample. This approach simplifies the presentation of the results and ensures that the same sample is used throughout its entire section of the results; otherwise, it would be confusing.
- SO5 variables: here, given the manual nature of the coding process, it is necessary to proceed using the sample.

During the coding process of the SO3 and SO5 variables, 100% of the sample was independently manually supervised or coded by the three researchers who authored this study to ensure reliability. Inter-coder reliability was assessed using Cohen's Kappa, with coefficients of 0.93 and 0.85, respectively, indicating strong agreement for the variables of both SOs. In cases of discrepancy, issues were resolved by majority or consensus. Notably, the first coefficient supports the analytical procedure for the SO3 variables using the representative sample, as it allows quantifying the differences between automatic coding without supervision and coding with supervision, the latter scenario being considerably more representative.

3.3. Data analysis

Given that the collected data has been systematized into variables that facilitate numerical analysis, statistical analysis is employed as the primary technique for data analysis. The instrumentation involves a series of procedures and tools that are detailed below.

Generally, in the different sections of the results, descriptive analysis is used to offer a general overview of each variable. That is, each variable is explained in detail before making comparisons and correlations.

A logarithmic adjustment is also used to model the relationships between variables. This adjustment is based on the logarithmic transformation of a variable and is particularly effective for representing relationships in which the growth or decline of the data follows a constant proportion. It is used to explore trends in the temporal evolution of the publications mathematically. To evaluate the accuracy of the fit, the mean squared error (MSE) is calculated, which indicates the average of the squared differences between the observed data and the predictions. The interpretation of the MSE also requires calculating the variance, which provides additional context regarding the dispersion of the errors.

In relation to the frequent terms in the news articles, three specialized tools for term frequency analysis are employed. First, MeaningCloud software is used, which allows the identification of the most frequent terms through its analysis function. Second, to generate visual representations of the data, the R programming language and its *wordcloud* library is used. Third, to view indicatively how terms are distributed across the main topics present in the news articles, the *BERTopic* library in Python is employed.

On the other hand, to explore the relationships between categorical variables, the chi-square test is employed. This technique allows for assessing whether there is a significant association between the categories by comparing the observed frequencies with the expected ones. In the present analysis, the test is used to examine how the website of publication, the categories of interests addressed, and the sentiment expressed in the content are interrelated.

Finally, a binary logistic regression (BLR) model is employed to analyze the probability of an event occurring or not within a binary variable, using multiple potential explanatory variables (Peng et al., 2002). This tool allows for determining whether the cause of publication of a news article can be linked to an indirect lobbying strategy or not. All research variables are included in order to determine which ones truly have explanatory power.

4. Results

As mentioned, the results are organized into five sections, each of which aligns with one of the five SO of the study.

4.1. Temporal and website distribution of the news articles

Between January 2013 and December 2023, a total of 13,431 news articles related to lobbies were published on the websites of the seven generalist digital newspapers, averaging 1,221 news articles per year. The website with the highest number of publications is elpais.com (n=3,091; 23.01%), followed by lavanguardia.com (n=3,052; 22.72%) and elmundo.es (n=2,413; 17.97%). The presence of news articles on the three most active websites accounts for 63.7% of the total, which is equivalent to 8,556 articles.

Variations are also observed based on the newspaper's ideology. The ones associated with left-wing ideology represent 50.33% (n=6,760) of the news articles, while those aligned with right-wing ideology account for 38.22% (n=5,134).

The trend over time in the number of publications shows a gradual decline, identifiable in three phases. Initially, between 2013 and 2016, there is a sharp drop in the total number of news articles published, with an average annual reduction of 17.90%. Later, from 2016 to 2019, the number of publications per year stabilizes, remaining around 1,100. In the final stage, after experiencing another decline in 2020, there is a recovery with an average annual growth of 12.31% between 2020 and 2023. Despite this recent recovery trend, the average year-on-year difference stands at -5.22%, with 2013 marking the highest value (n=2,100; 15.64%) and 2020 the lowest (n=746; 5.55%). The overall evolution can be observed in **Figure 1**.

[Figure 1]

(LEGEND) Figure 1: Distribution of news articles over time and by website of publication.

Equation (1), which represents the logarithmic adjustment that describe the temporal evolution, is:

$$(1) y = -533,09 \log x + 2069,20$$

The obtained MSE is 19,637.44, while the recorded variance is 163,068.36. These relatively low values suggest that, after calculating the coefficient of determination, the error in the logarithmic prediction does not constitute a significant proportion of the total data analyzed. This is evidenced by comparing the ratio of the MSE to the variance, which is 0.12.

4.2. Frequent terms used in the news articles

The average length of the news articles is 5,170.21 characters, which translates to 831.96 words per each one. Most of them fall within a length range of 2,000 to 3,000 characters. However, since the frequency distribution is asymmetric, leaning to the right, the average word count per news article is higher.

The global analysis of terms frequency, which is graphically shown in **Figure 2**, demonstrates a clear focus on European international politics, with "EU" and "Brussels" being particularly prevalent. Moreover, while terms related to politics and economics are dominant, the presence of some linked to more specialized areas is noteworthy.

[Figure 2]

(LEGEND) Figure 2: Word cloud featuring the most frequent terms used in the news articles.

To delve deeper into the frequent used terms, the Python library *BERTopic* is used. The program automatically determines that the most explanatory way to group the content of the news articles is into 339 topics. As it is practically impossible to interpret such a large number of topics, this presentation of topics is made solely as a general and indicative overview.

The main ones tend to focus on political issues. In this sense, and in line with the global analysis, the term groupings concentrate on three axes: a European one, including "Brussels", "EU", "Europe", "Europeans", "summit", and "Eurogroup"; an international one centered on the United States, with terms such as "Biden", "pressure", "Trump", "US", "power", "influence", and "millions"; and a Spanish national axis, which groups "politics", "government", "Rajoy", "Sánchez", "Madrid", "president", and "Spain".

Another prominent group of topics concentrate on economic and business matters, including: "company", "economy", "entrepreneurs", "investment", and "reform", and "plan". Partially related to this are the tourism-related topics, which comprise terms like "tourism", "development", "hotels", "travel", "vacation", "record", and "sector"; as well as a few topics related to environment and energy, featuring terms such as "energy", "future", "wind", "companies", "crisis", "plan", and "field".

From a more social perspective, several topics stand out that include a set of terms related to feminism and the LGTBIQ+ movement, grouping the following: "women", "work", "LGBT", "people", "equality", and "everyone", among others. A topic specifically focused on animal welfare also stands out, encompassing "animal", "reform", "welfare", "dogs", "law", and "years".

Regarding the images and videos included in the news articles, since they are published online, almost all of them feature visual elements to enhance the content (n=13,406; 99.81%).

4.3. Categories of interests addressed in the news articles

From this point onward, the data analysis is based on the statistically representative random sample of news articles, as outlined.

The classification into macro-categories of interests reveals a predominance of publications related to political-economic interests, representing 61.84% (n=235) of the total. News articles focused on social interests constitutes 25.79% (n=98), while those covering mixed interests account for 12.37% (n=47).

This concentration of news articles addressing political-economic interests is reflected in the fact that the two categories of interests with the highest number of publications are "Politics and foreign affairs" (n=118; 31.05%) and "Economy, business, and trade" (n=93; 24.47%). Additionally, there is a frequency exceeding 10% in the categories of "Justice and rights" (n=42; 11.05%) and "Environment and energy" (n=41; 10.79%). The general distribution of news articles by categories of interest can be seen in **Table 2**.

[Table 2]

Table 2: Categories of interests addressed in the news articles.

ID	Category of interests	News articles	%	Ideology
01	Politics and foreign affairs	118	31.05	Conservative
02	Economy, business, and commerce	93	24.47	Conservative
03	Society and social affairs	33	8.68	Progressive
04	Research and innovation	6	1.58	Conservative
05	Environment and energy	41	10.79	Progressive
06	Justice and rights	42	11.05	Progressive
07	Health and food safety	11	2.89	Progressive
08	Development and cooperation	6	1.58	Conservative
09	Primary sector	6	1.58	Progressive
10	Governance and administration	24	6.32	Progressive
-	Total	380	100	-

Note: the right column represents the ideology of the newspapers with the highest proportion of news articles on each category of interests. The analysis is conducted comparatively based on the overall relative cumulative distribution of the newspapers corresponding to each ideology.

The right column of the table requires detailed analysis. It is based on the percentage of news articles per category of interests, based on the newspaper's ideology. In this sense, the greatest absolute difference is observed in the category "Politics and foreign affairs". Although this category of interests is the most common in websites of both ideological currents, in right-wing ones, nearly 40% (39.21%) of the news articles are included on it, which exceeds the percentage observed in left-wing websites by 12.08%. In contrast, the most notable discrepancy in relative terms is found in the social categories of interests, where left-wing websites publish almost three times more news articles than their right-wing counterparts (10.78% compared to 3.92%), highlighting a marked difference in content prioritization (see **Figure 3**).

[Figure 3]

(LEGEND) Figure 3: Categories of interests addressed in the news articles based on the newspaper's ideology.

Advancing in the analysis of this phenomenon, a chi-square independence test is conducted to determine the relationship between two categorical variables: the website of publication and the category of interests. The obtained results (chi-square=68.32; p_value=0.090; df=54) indicate values close to the significance threshold, suggesting a potential relationship between the variables. However, these do not reach the necessary threshold to be considered statistically significant.

4.4. Sentiment expressed in the content of the news articles

Across the news articles in the sample, the average sentiment score is -11.18, on the scale from -100 to 100. This negative trend is reflected in the overall distribution of news articles across the five sentiment categories, where the imbalance is evident: 5.53% (n=21) of them present a very negative sentiment; 42.10% (n=160), negative; 23.67% (n=90), neutral; 26.84% (n=102), positive; and 1.83% (n=7), very positive. Moreover, it is observed that an overwhelming majority are written subjectively (n=346; 91.05%) and in a non-ironic tone (n=878; 99.47%). The use of objective language is uncommon (n=34; 8.95%), and the use of irony is almost nonexistent (n=2; 0.53%).

In six out of the seven websites, the average sentiment score is below 0. The three lowest scores correspond to elcorreo.com (average=-35.26), elpais.com (average=-30.30), and 20minutos.es (average=-27.67). The

exception is lavanguardia.com, which has an average sentiment score of 5.98. Furthermore, when examining the average sentiment score according to the newspaper's ideology, it is observed that left-leaning ones have an average of -7.16; center, -27.67; and right-leaning, -20.32.

A comparative analysis, shown in **Figure 4**, is also conducted based on the categories of interests addressed in the news articles. Only three categories show a positive sentiment score: "Economy, business, and commerce", "Research and innovation", and "Environment and energy", with the highest average score being 6.17 and the median at 5. The remaining seven categories record negative sentiment scores, with particularly low extremes in the categories of interests "Primary sector", which has an average sentiment score of -37.34; "Development and cooperation", with -25.67; and "Politics and foreign affairs", with -20.17.

[Figure 4]

(LEGEND) Figure 4: Boxplot of the sentiment expressed in the content of the news articles based on the categories of interests addressed.

The coherence in negative sentiment is also evident when examining variations according to macro-categories of interests. The average sentiment score in news articles addressing political-economic interests is -10.98, while in articles addressing mixed or social interests, the average is -11.48.

Two additional chi-square independence tests are conducted to explore the potential correlation between the sentiment and two other categorical variables: the website of publication and the category of interests. On one hand, the results of the first test (chi-square=44.97; p_value=0.005; df=24) reveal a statistically significant relationship between the website of publication and the sentiment. As previously noted, the sentiment score is negative in six of the seven evaluated websites, with lavanguardia.com being the only one that deviates from the trend and, consequently, the main cause of the observed dependency result. To verify the actual influence of lavanguardia.com, an additional independence test is conducted excluding its data. In this case, the results obtained (chi-square=16.77; p_value=0.668; df=20) unequivocally suggest independence, thus emphasizing the determining role of the values of the Catalan newspaper. On the other hand, the analysis of the relationship between the category of interests and the sentiment suggests independence (chi-square=44.17; p_value=0.164; df=36). The resulting p_value from the test implies an 83.6% probability that any detected association is not casual.

4.5. Influence of indirect lobbying strategies in the publication of the news articles

The analysis of the results from the sample shows that 34.21% (n=130) of the news articles about lobbies on the websites of generalist media newspapers are published as a result of indirect lobbying strategies. In **Table 3**, the differences in the cause of publication are presented based on the rest of the variables.

[Table 3]

Table 3: Cause of publication of the news articles based on contrast variables.

Cause of publication (%)	Indirect lobbying strategy	Autonomous decision	*
Contrast variable: Website of publication			
elpais.com	16.05 (13 news articles)	83.95 (68 news articles)	
elmundo.es	20.45 (9 news articles)	79.55 (35 news articles)	
lavanguardia.com	50.00 (77 news articles)	50.00 (77 news articles)	***
abc.es	35.90 (14 news articles)	64.10 (25 news articles)	
20minutos.es	22.22 (2 news articles)	77.78 (7 news articles)	

elcorreo.com	15.79 (3 news articles)	84.21 (16 news articles)	
elperiodico.com	35.29 (12 news articles)	64.71 (22 news articles)	*
Contrast variable: Newspaper's ideology			
Left	37.92 (102 news articles)	62.08 (167 news articles)	*
Centre	22.22 (2 news articles)	77.78 (7 news articles)	
Right	25.49 (26 news articles)	74.51 (76 news articles)	
Contrast variable: Macro-category of interests			
Political-Economic	30.64 (72 news articles)	69.36 (163 news articles)	
Mixed	63.83 (30 news articles)	36.17 (17 news articles)	***
Social	28.57 (28 news articles)	71.43 (70 news articles)	
Contrast variable: Category of interests			
Politics and foreign affairs	11.01 (13 news articles)	88.99 (105 news articles)	
Economy, business, and commerce	53.76 (50 news articles)	46.24 (43 news articles)	***
Society and social affairs	39.39 (13 news articles)	60.61 (20 news articles)	**
Research and innovation	33.33 (2 news articles)	66.67 (4 news articles)	
Environment and energy	63.41 (26 news articles)	36.59 (15 news articles)	***
Justice and rights	21.43 (9 news articles)	78.57 (33 news articles)	
Health and food safety	27.27 (3 news articles)	72.73 (8 news articles)	
Development and cooperation	16.67 (1 news article)	83.33 (5 news articles)	
Primary sector	66.67 (4 news articles)	33.33 (2 news articles)	***
Governance and administration	37.50 (9 news articles)	62.50 (15 news articles)	*
Contrast variable: Sentiment			
(-) N. Very negative	4.76 (1 news article)	95.24 (20 news articles)	
N. Negative	17.50 (28 news articles)	82.50 (132 news articles)	
NEU. Neutral	37.78 (34 news articles)	62.22 (56 news articles)	*
P. Positive	61.76 (63 news articles)	38.24 (39 news articles)	***
P (+). Very positive	57.14 (4 news articles)	42.86 (3 news articles)	***
Note: the right column represents the variation from the overall average percentage of news articles published as a result of an indirect lobbying strategy. * (Between 0.01% and 5.00% above the average). ** (Between 5.01% and 10.00% above the average). *** (More than 10.01% above the average).			

As can be observed based on the website of publication, the value for lavanguardia.com particularly stands out. When performing a comparative analysis based on the newspaper's ideology, this is balanced by the data from elpais.com, which shows a reduced influence of indirect lobbying strategies. The categories of interests with the highest proportion of news articles published as a result of indirect lobbying strategies are "Primary sector", "Environment and energy", and "Economy, business, and trade". Two of these categories represent mixed interests, resulting in the corresponding macro-category showing a proportionally high value. Ultimately, a clear trend is observed regarding the sentiment: the more positive the sentiment expressed in the content, the higher the likelihood that its publication was the result of an indirect lobbying strategy.

To deepen the understanding of how these variables can explain the cause of publication, a BLR model is implemented. The model aims to identify which ones have explanatory power regarding the cause of publication of the news article, whether it results from an indirect lobbying strategy or is autonomously published by the newspaper.

The first step is to identify the relevant variables. Among all variables included in the study, the sentiment, the category of interests, and the website of publication are considered relevant for the model. To ensure

that the model has statistical grounding, the Nagelkerke R-squared test, the Hosmer-Lemeshow test, and the omnibus test of the model coefficient are conducted. The results indicate that the regression explains 42% of the variability in the binary variable. Additionally, the result from the Hosmer-Lemeshow test ($p_value=0.863$) is close to 1, suggesting a very satisfactory fit of the model to the data.

The classification table is organized into three stages, each associated with the three relevant variables mentioned. Overall, as can be seen in **Table 4**, after the final stage, the BLR shows an overall explanatory power of 78.16%.

[Table 4]

Table 4: Classification table of the BLR model.

Step	Variables	Observed		Predicted		
				Indirect lobbying strategy		Correct %
				0	1	
Step 1	Sentiment	Indirect lobbying strategy	0	217	33	86.80
			1	73	57	43.85
		Percentage				72.11
Step 2	Category of interests Sentiment	Indirect lobbying strategy	0	211	39	84.40
			1	56	74	56.92
		Percentage				75.00
Step 3	Website of publication Category of interests Sentiment	Indirect lobbying strategy	0	214	36	85.60
			1	47	83	63.85
		Global percentage				78.16

5. Discussion

In general, the media coverage of lobbies in the Spanish digital generalist press highlights the importance of these actors in the media environment (Moreno-Cabanillas et al., 2024; Scott, 2018). Regarding the temporal evolution of coverage, there are two critical points. The first peak occurs in 2013, marking the highest value, which should be contextualized within the economic crisis that began in 2008 and intensified lobbying activities (Castillo-Esparcia et al., 2017). The second happens in 2020, when a sharp decline is observed due to the concentration of publications on pandemic-related topics, which completely altered the media communication landscape (Gong and Firdaus, 2024; Wicke and Bolognesi, 2021).

The analysis of temporal fluctuations is complemented by a comparison of media coverage based on the publication website of publication. The distribution is relatively balanced, although those with higher audience ratings (AIMC, 2024) and notoriety (Serna-Ortega et al., 2024) show the highest volumes of news articles. Moreno-Cabanillas et al. (2024) emphasize this balanced distribution, also highlighting a more significant presence of these actors in generalist newspapers compared to specialized ones. This finding is partially reflected in Mitchell et al.'s (2014) study, which notes that digital news consumption tends to be uniform, regardless of consumers' ideological inclinations.

Regarding the terms most frequently used in the news articles, the results show a strong connection to both national and international political issues. Lobbies, as intermediaries between society and decision-makers, influence policies and legislation by strategically utilizing the media (Walker, 2009; Woll, 2006). The prominent presence of terms related to the EU reflects the importance of supranational politics and the

growing influence of lobbies in Europe (Bernhagen and Mitchell, 2009; Colli and Adriaensen, 2020; Dür and Mateo, 2024).

When analyzing the categories of interests addressed in the news articles, a wide variety can be observed, which illustrates the organizational diversity within the operational scope of these entities (Lock and Davidson, 2024; Molins et al., 2016). This finding confirms the broad range of organizations grouped under the term "lobby" and underscores the extensive classification of these groups (Binderkrantz et al., 2017; Castillo, 2011).

In that sense, it is found that approximately two-thirds of the news articles focus on political-economic interests. This prioritization of certain interests within media agendas can be linked to agenda-setting (McCombs and Shaw, 1972) and agenda-building (Cobb and Elder, 1971) theories, and reflects the power of the media in shaping which topics are perceived as salient by the public. Although this issue has not been thoroughly studied in the Spanish context, Moreno-Cabanillas et al. (2024) find a high representation of employers' associations, public institutions, and companies. Internationally, according to Bernhagen and Trani (2012), lobbies with political or economic focuses tend to receive more media coverage without needing specific strategies, unlike social groups, which must concentrate on particular issues to gain attention. De Bruycker and Beyers (2015) add that, in the EU, the visibility of lobbies depends more on their stance regarding legislation than on their type or interest, as the media tends to focus on opponents and negative-tone news (Kätsyri et al., 2016).

To further explore the distribution of news articles by categories of interests, data is compared based on the website of publication. Right-wing ones show a higher proportion of political-economic news articles. Previous studies support this idea. Ejalde et al. (2018) observed that conservative media focus more on political news, while progressive media tend to cover social issues. Moreno-Cabanillas et al. (2024) confirmed that *elpais.com*, with a left-wing ideological orientation, covers more stories on gender rights and the LGBTQ+ community compared to right-wing media outlets like *abc.es* or *elmundo.es*. Entman (2007), on this topic, argues that media framing, priming, and agenda-setting are interconnected tools that structure political power and influence public perception. Ideological orientation acts as a lens through which topics are highlighted or downplayed, contributing to news slant and bias. Additionally, Stroud (2011) highlights that individuals often select media sources that match their own political beliefs, a phenomenon known as partisan selective exposure. Such behavior reinforces the influence of media ideology, as news consumers interpret content through their own perceptual lens, potentially contributing to a more polarized public and creating a feedback loop that influences both media production and consumption.

The evaluation of the sentiment expressed in the content shows that most news articles convey a negative sentiment. This negative focus may reflect the public perception of lobbies as opaque entities attempting to dishonestly influence regulatory processes (Oliver-González, 2018; Potters and Van Winden, 1992). However, this tendency can also be understood in light of the well-documented negativity bias in media coverage, which suggests that negative information attracts more public attention and engagement (Soroka, 2012; Trussler and Soroka, 2014). Regardless of the categories of interests addressed or the website of publication, negative sentiment remains consistent. The only notable exception is *lavanguardia.com*, which stands out for its more positive sentiment. Various studies on the media treatment of social and political issues (Amenta et al., 2017; Boomgaarden et al., 2011; Robinson, 2001) have found mixed conclusions, underscoring the importance of contextual and thematic differences.

Finally, it is found that more than a third of the news articles are published as a result of an indirect lobbying strategy. This supports the assertions of Westin (2003) and Castillo (2011), who, along with other researchers, have pointed out the importance that lobbies place on media visibility, highlighting that their demands gain political relevance through media exposure (Neal, 2008). Viewed from another perspective,

this idea can be linked to the notion that the media have the potential to act as gatekeepers or even as lobbies themselves (Trapp and Laursen, 2017). This also raises questions about how independent media outlets truly are and to what extent their agendas are shaped by external actors (Herman and Chomsky, 2008). Again, this can be linked to agenda-setting theory (McCombs and Shaw, 1972), going beyond the simple question of which issues receive attention and focusing on the way they are presented, a process through which the media exert power in shaping public understanding and influencing policy debates (Entman, 2007).

The comparison of proportions by website of publication based on the cause of publication shows that *lavanguardia.com* exhibits particularly high values of news articles published as a result of an indirect lobbying strategy. This may be associated with the character of the Catalan newspaper, whose editorial line has historically shown proximity to certain regional political and economic interest groups. Notably, *lavanguardia.com* also registers the most positive sentiment among all the analyzed websites. In turn, the relation is reinforced when examining the proportional distribution of articles resulting from lobbying communication strategies across different sentiment ranges categories. This pattern is further reflected in the model synthesizing the explanatory power of all variables on the cause of publication, where sentiment acts as a relevant variable alongside the categories of interests and the website of publication.

The explanatory capacity of the relevant variables in the model lies in their theoretical and empirical connection with the mechanisms of lobbying communication strategies. The sentiment variable captures the emotional framing of content (Gorostiza-Cerviño et al., 2024), which is directly linked to how indirect lobbying strategies seek to influence the tone and perception of their activities in the media. The variable related to the categories of interests, being a pertinent contrast variable (Serna-Ortega et al., 2025), reflects the thematic orientation of the news, providing insight into which types of interests are more likely to be associated with indirect lobbying strategies. This, in part, helps to address the gap in the scientific literature (Baumgartner and Leech, 2001; De Bruycker and Beyers, 2015; Moreno-Cabanillas et al., 2024) highlighted in the theoretical framework regarding the causes of variations in media coverage according to interests, noting that one possible cause may be the greater implementation of indirect lobbying strategies. Finally, the inclusion of the website of publication variable allows the model to account for potential editorial patterns that may condition how lobbying-related information is presented (Moreno-Cabanillas et al., 2024). Although its inclusion could be interpreted as overfitting, the model already demonstrated high explanatory power prior to this addition, so it does not pose a concern.

6. Conclusions

The longitudinal analysis conducted, has allowed for the evaluation of the media coverage of lobbies in Spanish generalist digital press. The significance of the following conclusions lies in recognizing media coverage as a crucial conditioning dimension on the influence capacity of these organizations.

In summary, the study reveals distinct temporal trends, with peak coverage in 2013 driven by the economic crisis and a sharp decline in 2020 due to pandemic-related news dominance. The diverse categories of interests addressed in the news articles reflect the broad organizational scope of lobbies, with a notable emphasis on political-economic interests. Additionally, sentiment analysis indicates a negative portrayal of lobbies, which may be linked to the public perceptions of their influence on regulatory processes. The findings also highlight the influence of indirect lobbying strategies in media coverage, which are also associated with a positive sentiment expressed in the content of the news articles.

These findings have practical implications for communication professionals and lobbyists, as well as for the understanding of democratic processes. The connection between indirect lobbying strategies and positive sentiment shows that proactive communication can shape how organizations are portrayed in the media. Professionals can use this insight to present their work in a clear and constructive way, making it

easier for journalists and the public to understand. Similarly, lobbyists can learn which approaches are most likely to gain media coverage, which adds practical value. At the same time, there are important implications for democracy and transparency, as the outlined connection can influence public accountability and the diversity of information available to citizens. If some interests consistently get more favorable attention, it may limit the range of perspectives presented in social debates. Being aware of these dynamics help to improve professional communication practices and also reflect on how media and lobbying interactions affect fair representation and trust in democratic processes.

To appropriately assess this information, it is necessary to consider a series of limitations present in the study. The collection of news presents three limitations. Although the seven main national newspapers have been included, some local or regional newspapers may have greater influence and readership in certain areas of Spain, potentially leading to an incomplete representation of those regions. The selection also does not include native digital media, which limits the variety of sources analyzed. In addition, the collection of news articles relies on the presence of specific terms within the content. While this approach is necessary to ensure that the dataset remains relevant to the topic under study, it may introduce certain biases. Concerning the limitations associated with the data systematizing and analyzing procedures, two factors should be taken into account. The study period coincides with a time of rapid technological change and evolving media consumption patterns, which may have influenced both the form and scope of the publications. It is also important to acknowledge that the research focuses exclusively on the Spanish context, limiting the generalizability of the findings at the international level.

Taking into account the exposed limitations, several directions for future research are suggested. First, conducting a study focused on native digital newspapers would be valuable, as it would allow for a comparison of the results obtained and an analysis of differences in coverage. Additionally, especially considering the results from lavanguardia.com, analyzing the websites of local or regional newspapers is particularly relevant, as they can complement the findings. Another line of research could qualitatively explore the relationship between the newspaper's ideology and the treatment of lobbies. Besides that, given that lavanguardia.com exhibits a much more positive sentiment and a significantly higher rate of articles published as a result of indirect lobbying strategies compared to the average, a case study could provide valuable insights into the underlying causes. Additionally, investigating the specific communication actions that are most effective in obtaining media coverage could enrich the field of study on the interaction between lobbies and the media.

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9. Author contributions

Á.S.-O., A.M.-C., and E.C.-O. contributed to the conception and design of the study. Á.S.-O. collected the data. Á.S.-O., A.M.-C., and E.C.-O. performed the analysis. A.M.-C., and E.C.-O. interpreted the results and wrote the manuscript. All authors contributed equally to this work. All authors reviewed and approved the final version of the manuscript.

10. Competing interests

The authors declare no competing interests.

11. Ethical approval

This study does not contain any studies with human participants performed by any of the authors.

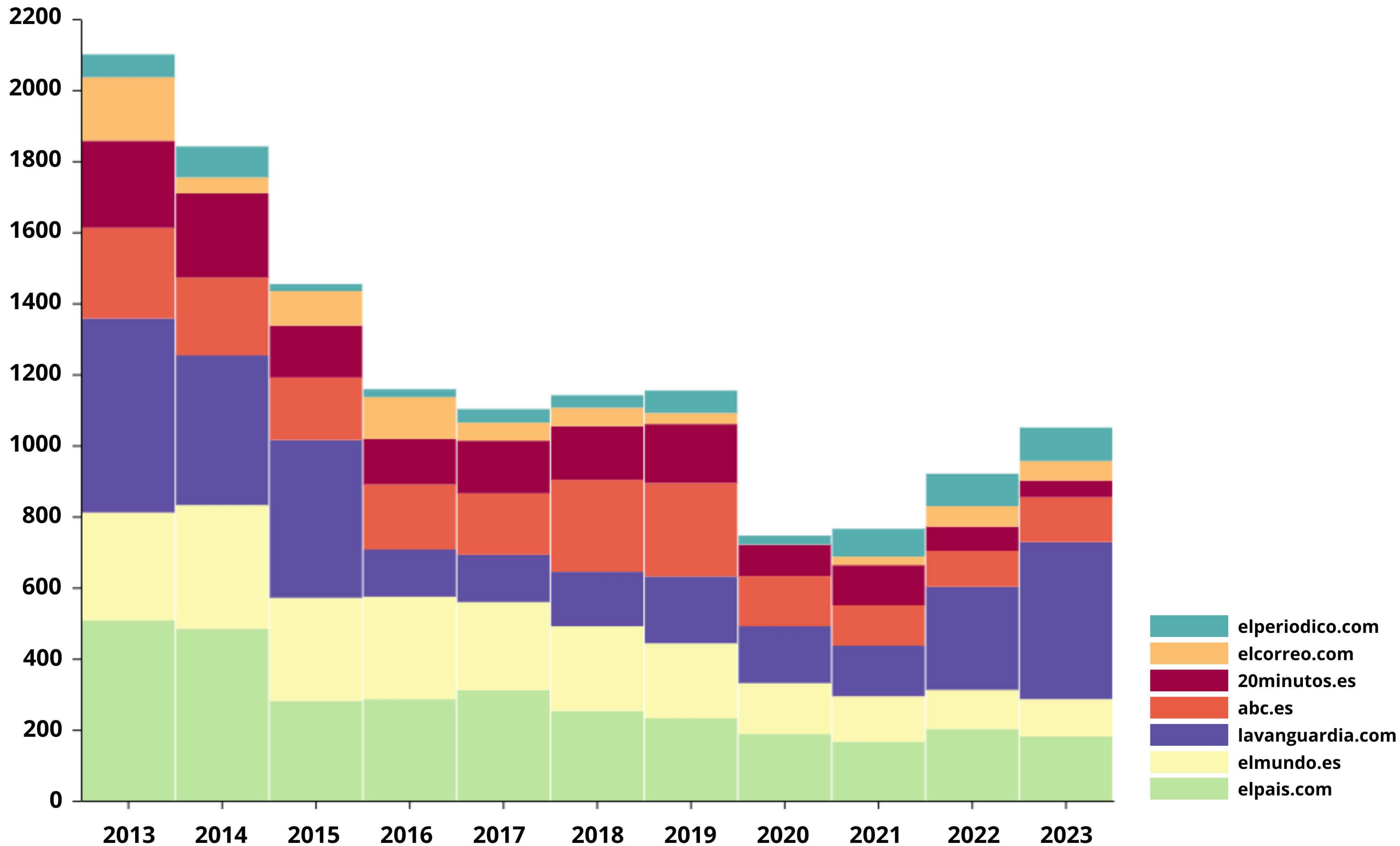
12. Informed consent

This study does not contain any studies with human participants performed by any of the authors.

13. Data availability

The dataset for the present study is available on Harvard Dataverse at <https://doi.org/10.7910/DVN/0XXFQI> (full link: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0XXFQI>)

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